UFSM00066 · Sistemas Dinâmicos
Lista de exercícios · Transformada de LaPlace
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3. Encontre $F(s)$ ou $f(t)$ como indicado.

a) $\mathcal{L}\{te^{10t}\}$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \rightarrow \frac{1}{s-10} \rightarrow -\frac{d}{ds}\left(\frac{1}{s-10}\right) = \frac{1}{(s-10)^2}$$

b) $\mathcal{L}\{e^{-2t}cos(4t)\}$

$$\mathcal{L}\{e^{at}\cos(\omega t) = \frac{s-a}{(s-a)^2+\omega^2} \rightarrow \frac{s+2}{(s+2)^2+4^2}$$

c) $\mathcal{L}\{t(e^t + e^{2t})^2\}$

$$(e^t+e^{2t})^2 = e^{2t} + 2e^{3t} + e^{4t}$$

$$f(s) \rightarrow \frac{1}{s-2} + \frac{2}{s-3} + \frac{1}{s-4} \rightarrow F(s) = \frac{-d f(s)}{ds} = \frac{1}{(s-2)^2} + \frac{2}{(s-3)^2} + \frac{1}{(s-4)^2}$$

d) $\mathcal{L}\{e^{2t}(t-1)^2\}$ → Neste ponto eu encontrei a fórmula 5 na tabela e descobri que não preciso separar

$$e^{2t}(t^2-2t+1) = t^2e^{2t} - 2te^{2t} + e^{2t}$$

$$F(s) = \frac{2}{(s-2)^3} - \frac{2}{(s-2)^2} + \frac{1}{s-2}$$

e) $\mathcal{L}\{e^t cos^2(3t)\}$ $\rightarrow \cos^2(3t) = \frac{1+\cos(6t)}{2}$

$$\hookrightarrow \mathcal{L}\left\{\frac{e^t}{2}\right\} + \mathcal{L}\left\{\frac{e^t\cos(6t)}{2}\right\} = \frac{1}{2(s-1)} + \frac{s-1}{2((s-1)^2+36)}$$

f) $\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-a)^{n+1}}\right\} = \frac{t^n e^{at}}{n!}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\} = \frac{t^3e^t}{6}$$

g) $\mathcal{L}^{-1}\left\{\frac{s}{\underbrace{s^2+4s+5}_{(s-2)^2+1}}\right\}$

$$\frac{s}{(s-2)^2+1} = \frac{s-2+2}{(s-2)^2+1} = \frac{s-2}{(s-2)^2+1} + \frac{2}{(s-2)^2+1}$$

$$f(t) = e^{2t}\cos t + 2e^{2t}\sin t = e^{2t}(\cos t + 2\sin t)$$

h) $\mathcal{L}^{-1}\left\{\frac{(s+1)^2}{(s+2)^4}\right\}$

$$(s+1)^2 = (s+2-1)^2 = (s+2)^2 - 2(s+2) + 1$$

$$\frac{(s+1)^2}{(s+2)^4} = \frac{1}{(s+2)^2} - \frac{2}{(s+2)^3} + \frac{1}{(s+2)^4}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^n}\right\} = \frac{t^{n-1}}{(n-1)!}e^{-at}$$

$$f(t) = e^{-2t}\left(t - t^2 + \frac{t^3}{6}\right)$$

i) $\mathcal{L}\{e^{2-t}\mathcal{U}(t-2)\}$ $\rightarrow e^{2-t} = e^{-(t-2)}$; $f(t) = e^{-(t-2)}u(t-2)$

$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$

$\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$ $\rightarrow F(s) = \frac{e^{-2s}}{s+1}$

j) $\mathcal{L}\{cos(2t)\mathcal{U}(t-\pi)\}$ $\rightarrow \cos(2t) = \cos(2(t-\pi))$

$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}f(s)$

$F(s) = e^{-\pi s} \cdot \frac{s}{s^2+4}$

l) $\mathcal{L}\{(t-1)^3e^{t-1}\mathcal{U}(t-1)\}$

$f(t) = (t-1)^3 e^{t-1}$

$\mathcal{L}\{f(t)\} = \frac{6}{(s-1)^4}$ $F(s) = e^{-s}\frac{6}{(s-1)^4}$

m) $\mathcal{L}\{te^{t-5}\mathcal{U}(t-5)\}$

$t = (t-5)+5$

$f(t) = ((t-5)+5)e^{t-5}$

$F(s) = e^{-5s}\left(\frac{1}{(s-1)^2} + \frac{5}{s-1}\right)$

n) $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\}$

$\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t \rightarrow f(t) = \sin(t-\pi)u(t-\pi)$

o) $\mathcal{L}^{-1}\left\{\frac{se^{-\pi s/2}}{s^2+4}\right\}$

$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos(2t) \rightarrow f(t) = \cos\left(2\left(t-\frac{\pi}{2}\right)\right)u\left(t-\frac{\pi}{2}\right)$

p) $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\}$

$\frac{1}{s^2(s-1)} = \frac{-1}{s} + \frac{-1}{s^2} + \frac{1}{s-1}$

$f(t) = -1-t+e^t \rightarrow f(t) = (-1-(t-2)+e^{t-2})u(t-2) = (e^{t-2}-t+1)u(t-2)$

q) $\mathcal{L}\{te^{2t}sen(6t)\}$

$\mathcal{L}\{\sin(6t)\} = \frac{6}{s^2+36} \rightarrow F(s) = \frac{6}{(s-2)^2+36}$

$\mathcal{L}\{t f(t)\} = -\frac{d}{ds}F(s) \rightarrow F(s) = \frac{12(s-2)}{((s-2)^2+36)^2}$

r) $\mathcal{L}^{-1}\left\{\frac{s+1}{\underbrace{(s^2+2s+2)}_{(s+1)^2+1}{}^2}\right\}$

$\mathcal{L}\{te^{-t}\sin t\} = \frac{2(s+1)}{((s+1)^2+1)^2}$

$\hookrightarrow f(t) = \frac{1}{2}te^{-t}\sin t$